Inter-Frequency Offload

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1 Inter-Frequency Offload Overview

> The feature is part of load management. It is used to offload traffic to the E-UTRAN network if the traffic load in the E-UTRAN cells exceeds a configured level.

Access Type:	LTE
Feature Identity:	FAJ 121 3061
Value Package Name:	Inter-Vendor Load Management
Value Package Identity:	FAJ 801 0416
Node Type:	Baseband Radio Node
Licensing:	Licensed feature. One license required for each node.

Summary

The Inter-Frequency Offload feature is part of the Load Management features. Load balancing relations are configured between two E-UTRAN cells. The Inter-Frequency Offload feature is used to offload traffic to the E-UTRAN network (Ericsson or non-Ericsson E-UTRAN FDD or TDD cells) if the traffic load in the E-UTRAN cells exceeds a configured level.

The feature uses the function introduced by the Inter-Frequency Load Balancing feature for the assessment of traffic load based on the presence of E-RABs in the cell and the QoS class to which they belong.

This feature is an alternative feature of Inter-Frequency Load Balancing. The main difference is there is no load information exchange between the cells in the Inter-Frequency Offload feature.

The Inter-Frequency Offload feature provides the following benefits:

- Certain operators can use the spectrum and equipment in a more efficient way by pushing traffic between the carriers. This applies to operators that have agreements with other operators with overlaying carriers, and operators that have overlaying carriers operated by nodes of different vendors.
- It reduces the risk for a UE to be trapped in an overused E-UTRAN cell providing poor performance.
- It enhances the usage of all frequencies within a network.
- Inter-frequency offload can be applied in different network topologies where a large number of small inter-frequency cells exists, that are not feasible for load balancing. For example, heterogeneous network or small cell deployments.

Additional Information

More information about this feature and related topics can be found in the following documentation:

- 3GPP TS 36.413; E UTRAN S1 Application Protocol (S1AP)
- 3GPP TS 36.423, X2 Application Protocol (X2AP)

2 Dependencies of Inter-Frequency Offload

Features

Table 1 Feature Dependencies

Feature	Relationship	Description
At least one of these two features must be enabled:		
Coverage-Triggered Inter-Frequency Handover (FAJ 121 0877)	Prerequisite	The Coverage-Triggered Inter-Frequency Handover feature is required by the Inter-Frequency Offload feature to trigger handover based on inter-frequency offload while keeping TDD or FDD mode.
Intra-LTE Inter-Mode Handover (FAJ 121 3042)	Prerequisite	The Intra-LTE Inter-Mode Handover feature is required for the Inter-Frequency Offload feature to trigger handover based on inter-frequency offload while changing TDD or FDD mode.
Accelerated Inter-Frequency Load Balancing (FAJ 121 5036)	Related	The Accelerated Inter-Frequency Load feature makes this feature more configurable. At least one of the following features must be enabled for the Accelerated Inter-Frequency Load feature to operate: – Inter-Frequency Load Balancing – Inter-Frequency Offload – Inter-RAT Offload to WCDMA When uplink-heavy UE detection and exclusion are active in a cell, uplink-heavy UEs are excluded from load balancing mobility. This gives priority to such UEs for the mobility actions of the Uplink-Traffic-Triggered Mobility feature.
Admission-Triggered Offload (FAJ 121 3100)	Related	The Admission-Triggered Offload feature provides control of the load created by the GBR bearers in E-UTRAN cells. This is achieved by offloading the UEs to other LTE or WCDMA cells when GBR usage of the source cell exceeds the ATO threshold.
Automated Neighbor Relations (FAJ 121 0497)	Related	The Automated Neighbor Relations feature can automatically add or remove `EUtranCellRelation` MOs used by the Inter-Frequency Offload feature. When an `ExternalEUtranCellFDD` or `ExternalEUtranCellTDD` MO is created by the ANR, the attribute `lbEUtranCellOffloadCapacity` is set to `AnrFunctionEUtran.lbCellOffloadCapacityPolicy`. When an `EUtranCellRelation` MO is created by the ANR, the attribute `isHoAllowed` is set to `AnrFunctionEUtran.hoAllowedEutranPolicy`.
Basic Intelligent Connectivity (FAJ 121 4843)	Related	The Inter-Frequency Offload feature can be applied for EN-DC capable UEs with SN terminated bearers and EN-DC capable UEs without SN terminated bearers using the same criteria as for LTE UEs. The load balancing can be configured with the `EUtranCellFDD.lbActionForEndcUe` attribute or the `EUtranCellTDD.lbActionForEndcUe` attribute.
Basic Load Management (FAJ 121 3092)	Related	The objective for the load management functions in the eNodeB is to balance or offload load when possible. During the execution of load management functions the MP processor load can increase. To avoid affecting general traffic handling, the load management functions are temporarily paused if the MP load control state is higher than `NOT HIGH`.
Best Neighbor Relations for Intra-LTE Load Management (FAJ 121 3028)	Related	The Best Neighbor Relations for Intra-LTE Load Management feature provides the functions to find and evaluate suitable E-UTRAN cell relations for inter-frequency load balancing or inter-frequency offload automatically.
Carrier Aggregation-Aware IFLB (FAJ 121 3075)	Related	When the Carrier Aggregation-Aware IFLB feature is operable, more UE measurements are started. This can reduce the number of measurements that Inter-Frequency Offload can start. When uplink-heavy UE detection and exclusion are active in a cell, the UE might be moved to a neighbor cell before it is identified as uplink-heavy. If it is detected as uplink-heavy in the serving cell it was moved to, it might be immediately moved back to the source cell. After the last handover, the UE is identified as uplink-heavy and is not chosen for further load-balancing actions as long as it is marked as uplink-heavy.
Coverage-Adapted Load Management (FAJ 121 3077)	Related	The Coverage-Adapted Load Management feature optimizes the amount of load to be transferred to neighboring cells and the number of required UE measurements for load balancing purposes. It also allows handling large number of load balancing relations with better efficiency, which facilitates load balancing small cells.

Feature	Relationship	Description
Device-Based Frequency Filtering (FAJ 121 5154)	Related	The Device-Based Frequency Filtering feature is used to configure a list of unsupported frequencies for specific UEs. These frequencies are excluded from UE configuration messages sent by the eNodeB. Unsupported frequencies are configured using the `UnsupportedFreqProfile.unsupportedFreqList` attribute.
Inter-Frequency Load Balancing (FAJ 121 3009)	Related	The Inter-Frequency Load Balancing feature is an alternative to Inter-Frequency Offload. The main difference is that there is no load information exchange between the cells in Inter-Frequency Offload.
Inter-RAT Offload to WCDMA (FAJ 121 3048)	Related	Inter-Frequency Offload is a parallel feature with Inter-RAT Offload to WCDMA, which can be operated standalone or in combination.
Limited-Uplink-Aware IFLB (FAJ 121 4406)	Related	The Limited-Uplink-Aware IFLB feature affects this feature as cell subscription capacity is enhanced to consider the uplink capacity of the cell.
Multiple Frequency Band Indicators (FAJ 121 3054)	Related	The purpose of the Inter-frequency Offload feature is to offload traffic load above an offload threshold from an E-UTRAN cell to an Ericsson or non-Ericsson E-UTRAN FDD or TDD cell.
Service Specific Load Management (FAJ 121 3047)	Related	Service Specific Load Management enhances control of the UE selection process for Inter-Frequency Load Balancing, Inter-RAT Offload to WCDMA, and Inter-Frequency Offload.
Subscriber Triggered Mobility (FAJ 121 1788)	Related	If the RFPM function of Subscriber Triggered Mobility is activated and the UE has a SPID value associated with it, inter-frequency offload can be inhibited for that UE. It is inhibited only when the `FreqPrioEUTRA.loadBalancingAllowed` attribute related to the concerned `RATFreqPrio` MO for the concerned target carrier frequency is set to `FALSE`. When the Preferential Traffic Management solution is used, to inhibit inter-frequency offload for cell relations, the following conditions must be fulfilled: – `EUtranCellRelation.loadBalancing` must be set to `IFO_AND_IFLB`. – `FreqPrioEUTRA.offloadAllowed` must be set to `FALSE`
Load-Based Access Barring (FAJ 121 3093)	Conflicting	Subscription ratio measurements are disabled when automatic access class barring is triggered. In case of no subscription ratio measurements, outgoing Inter-Frequency Offload is disabled.
PLMN-Specific Access Barring (FAJ 121 4916)	Conflicting	Subscription ratio measurements are disabled when automatic access class barring is triggered. In case of no subscription ratio measurements, outgoing Inter-Frequency Offload is disabled.
Progressive Access Barring (FAJ 121 4746)	Conflicting	Subscription ratio measurements are disabled when automatic access class barring is triggered. In case of no subscription ratio measurements, outgoing Inter-Frequency Offload is disabled.
Mission-Critical Push-to-Talk (FAJ 121 4929)	Related	When the Mission-Critical Push-to-Talk feature is active, this feature treats mission-critical bearers like it treats VoLTE bearers. The recommended value for `QciProfilePredefined.qciSubscriptionQuanta` and `QciProfileOperatorDefined.qciSubscriptionQuanta` is 1 for GBR and non-GBR E-RABs with QCIs 65, 66, 69, and 70.
UE Throughput-Aware IFLB (FAJ 121 4219)	Related	UE Throughput-Aware IFLB is an enhancement to the Inter-Frequency Load Balancing feature. Throughput-aware UE selection estimates and compares possible UE throughput in the cellbefore using a UE for load balancing. A decision on whether to move the UE or not is based on the throughput difference. When uplink-heavy UE detection and exclusion are active in a cell, the UE might be moved to a neighbor cell before it is identified as uplink-heavy. If it is detected as uplink-heavy in the serving cell it was moved to, it might be immediately moved back to the source cell. After the last handover, the UE is identified as uplink-heavy and is not chosen for further load-balancing actions as long as it is marked as uplink-heavy.
Uplink-Traffic-Triggered Mobility (FAJ 121 5044)	Related	When the following features and the Uplink-Traffic-Triggered Mobility feature are enabled at the same time, after an uplink-heavy handover, the target eNodeB does not consider the UE for inter-frequency load balancing or offload decision until an internal timer runs out: – Inter-Frequency Load Balancing – Carrier Aggregation-Aware IFLB – Limited-Uplink-Aware IFLB

Feature	Relationship	Description
		– UE Throughput-Aware IFLB – Inter- Frequency Offload If uplink-heavy detection for load-balancing is active for any of these features, uplink-heavy UEs are excluded from load-balancing mobility in the cell.
Radio Resource Partitioning (FAJ 121 4571)	Related	The Inter-Frequency Offload feature is not partition-aware. This means that traffic is offloaded to E-UTRAN cells without considering the traffic load of the individual partitions. UEs from an overused partition might not be selected for offload. Performance might degrade for UEs that are offloaded to an overused partition in which resources are limited by the `ResourcePartitionGroupMember.resourcePartitionMaxShare` attribute.

Hardware

No specific hardware requirements.

Limitations

No limitations for this feature.

Network Requirements

The cell relations where inter-frequency offload is desired must be configured to allow load balancing actions and reporting, by setting `EUtranCellRelation.loadBalancing` attribute.

Note: Inter-frequency offload between cells on different nodes does not require X2 connection between the nodes. This is because inter-frequency offload operates without load information exchange.

System Functions

Table 2 System Function Dependencies

Function	Description
Inter-frequency measurements	The number of inter-frequency measurements increases when the Inter-Frequency Offload feature is activated. The increase depends on the number of inter-frequency handovers required to keep traffic load in balance.
Inter-frequency handover	The number of inter-frequency handovers increases when the Inter-Frequency Offload feature is activated.

3 Feature Operation of Inter-Frequency Offload

> The Inter-Frequency Offload feature is part of the Load Management features in LTE RAN. It is used to offload traffic to the E-UTRAN network if the traffic load in the E-UTRAN cells exceeds a configured level. Load balancing relations are configured between two E-UTRAN cells.

Operation Sequence Diagram

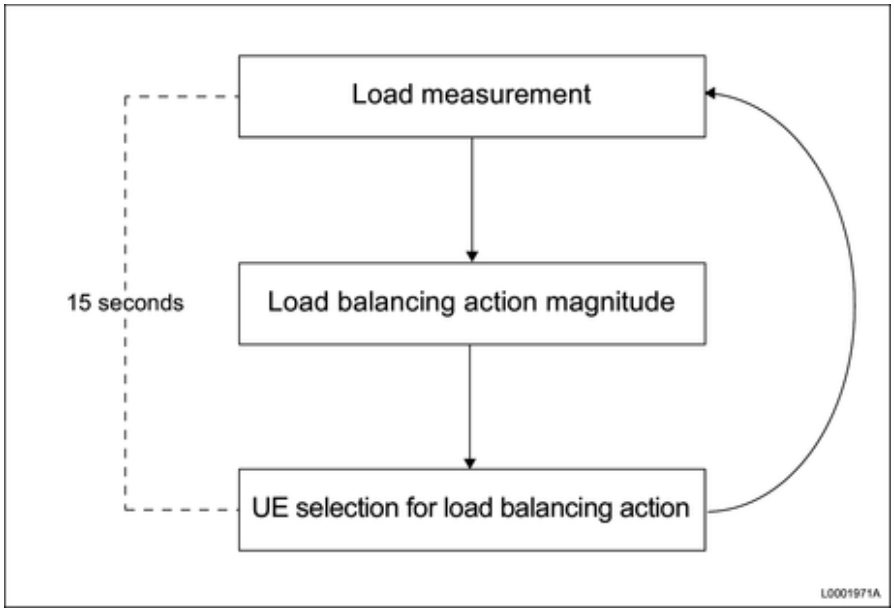

Figure 1 Inter-Frequency Offload

Process Steps

1. The traffic load in a cell is assessed repeatedly in certain intervals of a load balancing cycle.

   The current subscription ratio of the cell is calculated. The QCI subscription quanta values (`QciProfilePredefined.qciSubscriptionQuanta` and `QciProfileOperatorDefined.qciSubscriptionQuanta`) of all E-RABs (both GBR and non-GBR services) present in the cell are added. The sum is divided by the cell subscription capacity that is the value of `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` attributes.

   The subscription ratio determines further load balancing actions in the cell.

2. Based on the configured E-UTRAN trigger offload threshold, each source cell determines the amount of traffic load to be handed over to each target cell to balance the load differences.

   The configured E-UTRAN trigger offload threshold can be one of the following:

   – `EUtranCellFDD.lbEUtranTriggerOffloadThreshold`

   – `EUtranCellTDD.lbEUtranTriggerOffloadThreshold`

   This step can be done based on the following::

   – The load amount is proportional to the load difference between the source cell subscription ratio and threshold `EUtranCellFDD.lbEUtranTriggerOffloadThreshold` or `EUtranCellTDD.lbEUtranTriggerOffloadThreshold`.

   – The load amount cannot exceed the maximum load difference in the source cell in one round of the load balancing cycle. The maximum load difference is shown by the `LoadBalancingFunction.lbCeiling` attribute.

   – The load amount is proportional to the offload capacity of the target cell, `EUtranCellFDD.lbEUtranCellOffloadCapacity` or `EUtranCellTDD.lbEUtranCellOffloadCapacity`.

   – The load amount is inversely proportional to the sum of the subscription capacity of the following:

- The source cell
- The possible target cells for inter-frequency load balancing, or inter-frequency offload, or both

The sum of the subscription capacity can be described by `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` attribute.The possible target cells are those to which the attribute `EUtranCellRelation.loadBalancing` is set to `ALLOWED`, `OFFLOAD`, or `IFO_AND_IFLB`.

If multiple target cells exist, the load difference is split between these cells.

Note: The excess load in the source cell is typically not offloaded in a single round of load balancing actions. To remove the excess load, several rounds are required. By increasing the offload capacity for the target cells with the `EUtranCellFDD.lbEUtranCellOffloadCapacity` or `EUtranCellTDD.lbEUtranCellOffloadCapacity` attribute, the number of required rounds is reduced.

3. The UEs that are suitable for offload are selected.

   When the Preferential Traffic Management solution is enabled, the following applies:

   Note: Enabling subscriber group-based UE selection for offload requires the following configuration:

   – Priority cells reference cell profiles where `PtmCellProfile.cellType` is set to `PRIORITY`.

   – `PtmFunction.ptmEnabled` is set to TRUE.

   UEs are selected for offload based on the priority of their subscriber group. Up to 12 subscriber groups can be defined. Offload candidate selection works only on cells that reference `PRIORITY` Preferential Traffic Management cell profiles.

   To define the order in which UEs are offloaded by Inter-Frequency Offload, there is a need to configure the `PtmIfoConfig.ptmSubscriberGroupRef` list in the referenced cell profile. Groups at the top of the list are offloaded earlier as they have lower priority.

   Note: SPID-based UE selection for Inter-Frequency Offload requires X2 connection between the nodes for cells on different nodes.

   For more information, see Preferential Traffic Management Guidelines.

   When uplink-heavy UE detection for the Inter-Frequency Offload feature is configured and a UE is marked as uplink-heavy, it is excluded from load-balancing mobility actions. For more information, see Uplink-Heavy UE Detection.

   In the UE selection process, the UEs perform inter-frequency Event A5 measurements on the target frequency. This ensures that the UE has acceptable coverage in a target cell. Only those UEs are selected which support the target frequency band and trigger an Event A5 measurement report on the target cell.

   Note: Event A5 is also enables load balancing actions for UEs that are configured for Carrier Aggregation.

   The maximum number of measurements started for load balancing purposes during a 15-second load balancing cycle is limited.

   Note: A UE in connected mode is selected to perform a measurement on a target frequency for load balancing purposes at most once for each load balancing cycle.

   In case all target cells for a frequency have a back-off timer running, no UE measurements are started for that frequency.

4. The execution of load balancing actions is performed by handover of the selected UEs to the target cells.

   The handover of UEs is performed using the following normal inter-frequency handover procedures:

   – X2 Handover Preparation

   – S1 Handover Preparation

   – RRC Connection Reconfiguration

   Handover to a target cell is inhibited in the following cases:

   – The `EUtranCellRelation.isHoAllowed` attribute for the target cell is set to `FALSE`.

   – The cell is listed as forbidden in a HRL provided for the UE. For more information on the support of HRL, see Shared

LTE RAN.

– The RFPM function of the Subscriber Triggered Mobility feature is used. The `FreqPrioEUTRA.loadBalancingAllowed` attribute related to the `RATFreqPrio` MO for the target carrier frequency is set to FALSE for matching UE SPID.

  Note: If the Preferential Traffic Management solution is used, to inhibit inter-frequency offload for load balancing relations is inhibited. It can be achieved by setting the `EUtranCellRelation.loadBalancing` attribute to `IFO_AND_IFLB`, and the `FreqPrioEUTRA.offloadAllowed` attribute must also be set to FALSE.

  Otherwise, if the `FreqPrioEUTRA.loadBalancingAllowed` attribute is set to TRUE or there is no matching UE SPID, load balancing to the target cell for the UE is allowed.

  For more information on the support of SPID use, see Subscriber Triggered Mobility.

Table 3 Handover Cause Value for Offload-Triggered Handover

Handover Type	Cause Value Setting	Standard
S1 handover	The cause value is set in the `HANDOVER REQUIRED` message according to the `LoadBalancingFunction.lbCauseCodeS1SourceTriggersOffload` attribute.	*3GPP TS 36.413; E UTRAN S1 Application Protocol (S1AP)*
X2 handover	The cause value is set in the `HANDOVER REQUEST` message according to the `LoadBalancingFunction.lbCauseCodeX2SourceTriggersOffload` attribute.	*3GPP TS 36.423; E UTRAN X2 Application Protocol (X2AP)*

5. A node acting as offload target checks incoming handovers for the handover cause value configured as offload with the following parameters:

   – `LoadBalancingFunction.lbCauseCodeS1TargetAcceptsOffload`

   – `LoadBalancingFunction.lbCauseCodeX2TargetAcceptsOffload`

   An incoming offload handover is accepted if the traffic load of the target cell is below the accept offload threshold `EUtranCellFDD.lbEUtranAcceptOffloadThreshold` or `EUtranCellTDD.lbEUtranAcceptOffloadThreshold`. Otherwise the handover is rejected.

6. If the target E-UTRAN cell rejects the load balancing action handover, the source cell temporarily removes the E-UTRAN cell from the set of offload target cells. The E-UTRAN cell is restored in the set of offload target cells after a configurable back-off time, `LoadBalancingFunction.lbEUtranOffloadBackoffTime`.

3.1 Load Balancing Cycle

> The load balancing cycle is the time interval between two successive evaluations of traffic load balancing status. It is the time frame for the actions attempting to restore the traffic load balance.

Load balancing between cells is done repeatedly in successive load balancing cycles of 15 seconds.

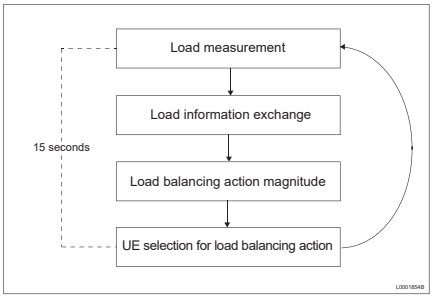

Figure 2 Load Balancing Cycle

Load Measurement	The load of each cell is estimated.
Load Information Exchange	Load information is exchanged between source and target cells. For inter-eNodeB load exchange, X2 messages are used.
Load Balancing Action Magnitude	The amount of offloading required to manage load between cells is determined. Higher loaded cells select some UEs as candidates for load balancing. The selected UEs are configured for event A5 inter-frequency measurement.
UE Selection for Load Balancing Action	The UEs suitable for offload are selected. The suitable UEs are those that reported to the source cell that they detected sufficient coverage in a target cell. Some suitable UEs are moved to detected target cells by inter-frequency handover.

3.2 Load Balancing Relation

> The load balancing relation is the logical relation between a load balancing source and a target cell. Load balancing relations are controlled with the `EUtranCellRelation.loadBalancing` attribute.

For load balancing features to steer traffic from one source cell to a target cell, a relation must be defined in the source cell. This relation is defined to the target cell using the `EUtranCellRelation` MO. The `EUtranCellRelation.loadBalancing` attribute controls whether load balancing, or offload, or both are allowed.

Note: Each defined load balancing relation is unidirectional.

3.3 GBR Traffic Load

GBR traffic is characterized by a relatively persistent E-RAB that generates a steady data stream. For example, an E-RAB for a voice call typically lasts for one or a few minutes. During the voice call, a stream of voice packets is transferred at a nearly constant rate.

Therefore, the number of E-RABs for GBR traffic present in a cell is a good basis for the assessment of the GBR traffic load.

3.4 Non-GBR Traffic Load

Non-GBR traffic is typically characterized by short but frequent data transactions. The individual data transactions can range from a few data packets to the transfer of large files. Many data transactions are completed within a short time, often less than a few seconds.

Non-GBR traffic has the following pattern:

1. The UE enters connected mode.
2. The E-RABs are established.
3. The data transaction, or a sequence of data transactions proceeds.
4. The UE is released after a period of inactivity.

This pattern is repeated, but each RRC connection and each E-RAB for the node is unique. The node is unable to correlate the different RRC connections generated by a UE.

Non-GBR traffic patterns are difficult to predict because they change often and quickly. The activity of an individual UE or E-RAB does not necessarily give much information about the activity a few seconds ahead. The number of E-RABs presented by the UEs to the LTE RAN can be used to assess the non-GBR traffic load generated by the UE groups. This applies for a UE group camped in a cell.

3.5 Subscription Ratio

> Subscription ratio is a relative traffic load measure. It is the ratio between the total QCI subscription quanta of all E-RAB instances currently configured in a cell and the cell subscription capacity of the cell.

It takes a value between zero and one for an under-subscribed cell. Values greater than one indicate an over-subscribed cell.

$$SubscriptionRatio = \frac{\sum qciSubscriptionQuanta}{cellSubscriptionCapacity}$$

Figure 3 Subscription Ratio

The configuration of the following values has to obtain the required load balancing behavior:

- `qciSubscriptionQuanta` (in MOs `QciProfilePredefined` or `QciProfileOperatorDefined`)
- `cellSubscriptionCapacity` (in MOs `EUtranCellFDD` or `EUtranCellTDD`)

This enables load balancing between cells with different UE groups, and cells with different traffic capacities.

Configuration of the assessment model offers control of the load balancing objectives in LTE RAN and opportunity for further changes.

The subscription ratio can be used to monitor the traffic load in a cell.

3.6 Cell Subscription Capacity

Cell subscription capacity is the nominal traffic handling capacity of a cell. Its value is an estimate of the total amount of QCI subscription quanta values the cell is expected to accommodate at full system load. Cell subscription capacity defines the target for load balancing to divide the traffic load between overlaid cells within the network. It can be configured with one of the following MOM attribute:

- `EUtranCellFDD.cellSubscriptionCapacity`
- `EUtranCellTDD.cellSubscriptionCapacity`

3.7 QCI Subscription Quanta

QCI subscription quanta is the representation of the anticipated or typical traffic handling capacity required for an E-RAB with certain QCI.

QCI subscription quanta is configured for a QoS class by setting the `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` attribute.

3.8 Offload Capacity

The offload capacity moderates the offload amount directed to the target E-UTRAN cell. This can be achieved with one of the following attributes:

- `EUtranCellFDD.lbEUtranCellOffloadCapacity`
- `EUtranCellTDD.lbEUtranCellOffloadCapacity`

The amount of offload to a specific target E-UTRAN cell is proportional to the `EUtranCellFDD.lbEUtranCellOffloadCapacity` or `EUtranCellTDD.lbEUtranCellOffloadCapacity` attribute of that target E-UTRAN cell.

Note: The excess load in the source cell is typically not offloaded in a single round of load balancing actions. To remove the excess load, several rounds are required. By increasing the offload capacity for the target cells with the `EUtranCellFDD.lbEUtranCellOffloadCapacity` or `EUtranCellTDD.lbEUtranCellOffloadCapacity` attribute, the number of required rounds is reduced.

3.9 Load Management and MP Load Control

The objective for the load management functions in the eNodeB is to balance or offload load when possible. However, the focus is on the air interface load. During the execution of load management functions the MP processor load may increase. To not affect general traffic handling, the load management functions are temporarily paused if the MP load control state is higher than `NOT HIGH`. For more information, see Basic Load Management.

3.10 Uplink-Heavy UE Detection

When uplink-heavy UE detection for the feature is configured, the UE selection process improves for load-balancing mobility actions. The feature excludes UEs from load-balancing mobility actions if they are marked as uplink-heavy. The uplink-heavy UE detection mechanism is applied to all UEs that connect to the LTE cell.

The information about the uplink-heavy state of the UE is stored in the UE context.

4 Parameters for Inter-Frequency Offload

Feature Configuration Parameters

- AnrFunctionEUtran.lbCellOffloadCapacityPolicy
- EUtranCellFDD.lbEUtranTriggerOffloadThreshold
- EUtranCellTDD.lbEUtranTriggerOffloadThreshold
- EUtranCellFDD.lbEUtranAcceptOffloadThreshold
- EUtranCellTDD.lbEUtranAcceptOffloadThreshold
- EUtranCellFDD.lbEUtranCellOffloadCapacity
- EUtranCellTDD.lbEUtranCellOffloadCapacity
- LoadBalancingFunction.lbCauseCodeS1SourceTriggersOffload
- LoadBalancingFunction.lbCauseCodeS1TargetAcceptsOffload
- LoadBalancingFunction.lbCauseCodeX2SourceTriggersOffload
- LoadBalancingFunction.lbCauseCodeX2TargetAcceptsOffload
- LoadBalancingFunction.lbEUtranOffloadBackoffTime

Affected Parameters

The implementation of this feature affects no parameters.

Parameters Affecting the Feature

Table 4 Parameters Affecting the Feature

Parameter	Description
EUtranCellRelation.loadBalancing	Offload is allowed only for cell relations where the EUtranCellRelation.loadBalancing attribute is set to OFFLOAD or IFO_AND_IFLB. **Note:** The IFO_AND_IFLB value can only be used if Preferential Traffic Management is enabled and configured.
ENodeBFunction.ulHeavyUeDetectTime	The attribute can be used to configure uplink-heavy UE detection.
ENodeBFunction.ulHeavyUeBsrThresh	The attribute can be used to configure uplink-heavy UE detection.
ENodeBFunction.ulHeavyUeBsrHyst	The attribute can be used to configure uplink-heavy UE detection.
LoadBalancingFunction.lbCeiling	The LoadBalancingFunction.lbCeiling attribute sets the maximum amount of load that can be offloaded in one round of load-balancing action.
FreqPrioEUTRA.loadBalancingAllowed	If the RFPM function of Subscriber Triggered Mobility is activated and the UE has a SPID value associated with it, inter-frequency offload can be inhibited for that UE. It is inhibited only when the FreqPrioEUTRA.loadBalancingAllowed attribute related to the concerned RATFreqPrio MO for the concerned target carrier frequency is set to FALSE. When the Preferential Traffic Management solution is used, to inhibit inter-frequency offload for cell relations, the following conditions must be fulfilled: – EUtranCellRelation.loadBalancing must be set to IFO_AND_IFLB. – FreqPrioEUTRA.offloadAllowed must be set to FALSE
FreqPrioEUTRA.offloadAllowed	
EUtranCellFDD.cellSubscriptionCapacity	The cell subscription capacity of the cell—EUtranCellFDD.cellSubscriptionCapacity or EUtranCellTDD.cellSubscriptionCapacity—affects the calculation of the amount of load to be offloaded.
EUtranCellTDD.cellSubscriptionCapacity	

Parameter	Description
EUtranCellFDD.lbUlHeavyUeDetectEnabled	The attribute can be used to enable uplink-heavy UE detection for load-balancing mobility actions.
EUtranCellTDD.lbUlHeavyUeDetectEnabled	The attribute can be used to enable uplink-heavy UE detection for load-balancing mobility actions.
PtmStmConfig.ptmSubscriberGroupRef	When the Preferential Traffic Management solution is enabled, the following applies. To define the order in which UEs are offloaded by Inter-Frequency Offload, there is a need to configure the PtmIfoConfig.ptmSubscriberGroupRef list in the referenced cell profile. Groups at the top of the list are offloaded earlier as they have lower priority.
PtmResOpUseConfig.ptmSubscriberGroupRef	

5 Network Impact of Inter-Frequency Offload

Capacity and Performance

The purpose of the Inter-Frequency Offload feature is to hand over traffic load that is above a configured threshold in E-UTRAN cells. Traffic load is handed over from overused E-UTRAN cells to a configured set of E-UTRAN cells with spare capacity. The ability to fully use the capacity on all the available and configured E-UTRAN cells is improved.

The feature enables traffic offload to other E-UTRAN cells if the traffic load in the own E-UTRAN cells exceeds a configured level. This increases the system capacity and reduces the risk for a UE to be trapped in an overused E-UTRAN cell.

The Inter-Frequency Offload feature enables to limit the amount of incoming handovers triggered by offload from another cell. The traffic load in the own cell can exceed the following thresholds:

- `EUtranCellFDD.lbEUtranAcceptOffloadThreshold`
- `EUtranCellTDD.lbEUtranAcceptOffloadThreshold`

In this case, the Handover Preparation is rejected if the Cause is set to the same value as the value of following attributes:

- For S1 handover, `LoadBalancingFunction.lbCauseCodeS1TargetAcceptsOffload`.
- For X2 handover, `LoadBalancingFunction.lbCauseCodeX2TargetAcceptsOffload`.

Mobility

The Inter-Frequency Offload feature redistributes users in connected mode from E-UTRAN cells to other E-UTRAN cells by traffic load triggered PS handover. This feature can increase the number of PS handovers. Handover is typically performed in good radio conditions.

The handover cause value for the Inter-Frequency Offload feature is set according to `LoadBalancingFunction.lbCauseCodeS1SourceTriggersOffload` and `LoadBalancingFunction.lbCauseCodeX2SourceTriggersOffload`.

The following counters are used to distinguish Inter-Frequency Offload handover from other types of handover:

- `EUtranCellRelation.pmHoPrepAttLteInterFLb`
- `EUtranCellRelation.pmHoPrepSuccLteInterFLb`
- `EUtranCellRelation.pmHoExeAttLteInterFLb`
- `EUtranCellRelation.pmHoExeSuccLteInterFLb`
- `EUtranFreqRelation.pmLbMeasuredUe`
- `EUtranCellRelation.pmLbQualifiedUe`

6 Performance of Inter-Frequency Offload

KPIs

The table lists the main KPIs associated with the feature, and the expected impact.

Table 5 Key Performance Indicators

KPI	Description
Added E-RAB Establishment Success Rate	When the Inter-Frequency Offload feature is activated, the accessibility success rate can slightly decrease for end-user services carried by E-RABs included in the E-RAB setup procedure.
Mobility Success Rate	The Mobility Success Rate KPI includes both the preparation of the target cell resources and the move from the source cell to the target cell.

Counters

- pmHoExeAttLteInterFLb
- pmHoExeAttNonMob
- pmHoExeSuccLteInterFLb
- pmHoExeSuccNonMob
- pmHoPrepAttLteInterFLb
- pmHoPrepAttNonMob
- pmHoPrepRejInOffloadAdmission
- pmHoPrepSuccLteInterFLb
- pmHoPrepSuccNonMob
- pmLbMeasuredUe
- pmLbQualifiedUe
- pmPtmIfoMeasuredUeDistr

Events

Table 6 Events - Inter-Frequency Offload

Event	Event Parameter	Description
INTERNAL_EVENT_LB_INTER_FREQ	EVENT_PARAM_NEIGHBOR_CGI	CGI of target cell participating in load balancing (if available)
	EVENT_PARAM_LB_DIFF	Load difference (in subscription ratio) between source and offload threshold
	EVENT_PARAM_LB_DIFF_INT_SIGN	Unsigned integer value of EVENT_PARAM_LB_DIFF; positive or negative
	EVENT_PARAM_LB_AMOUNT	Load balancing amount (in subscription quanta) to be moved from source to target cell
	EVENT_PARAM_LB_SOURCE_RATIO	Subscription ratio of source cell
	EVENT_PARAM_LB_TARGET_RATIO	Subscription ratio of target cell
	EVENT_PARAM_LB_TYPE	Type of Load Management relation applying for this target cell
INTERNAL_EVENT_UE_LB_MEAS	EVENT_PARAM_FREQ	Target frequency for which the UE is ordered to measure.
	EVENT_PARAM_TRIGGERING_FUNCTION	Indicates the function that has triggered the UE measurement and consequently the measurement report: – Load Balancing – Carrier Aggregation (CA) triggered

Event	Event Parameter	Description
		redirection – BNR for load balancing – ATO
	EVENT_PARAM_UE_CA_CAPABLE	Indicates whether the UE selected for load balancing related measurements is CA capable.
	EVENT_PARAM_NEIGHBOR_CGI	CGI of neighbor target cells for which the UE fulfilled the criteria for event A5.
	EVENT_PARAM_TRAVERSED_UES	Number of UEs checked and discarded before choosing this UE for load balancing related measurements.
INTERNAL_EVENT_UE_LB_QUAL	EVENT_PARAM_NEIGHBOR_CGI	CGI of neighbor target cells for which UE fulfilled event A5 criteria.
	EVENT_PARAM_HO_SOURCE_OR_TARGET_TYPE	Indicates the RAT of the source cell for incoming handover, or target cell for outgoing handover.
	EVENT_PARAM_LB_CANCELLED_REASON	Indicates the reason for not proceeding with a load balancing or CA triggered redirection action when a measurement report has been received from the UE: – CA criteria not met – Reported cell is not a valid load balancing target – Reported cell is a valid target, but `lbAmount` has already been satisfied for the cell – Service Specific Load Management (SSLM) related restrictions – RSRQ of target cell is below configured threshold – UE has signalled interest in MBMS service on serving frequency – UE has an ongoing VoIP call and does not support SRVCC – ATO related restrictions
	EVENT_PARAM_TRIGGERING_FUNCTION	Indicates the function that has triggered the UE measurement and consequently the measurement report: – Load balancing – CA triggered redirection – BNR for load balancing – ATO
INTERNAL_PROC_HO_EXEC_X2_OUT	EVENT_PARAM_PROC_HO_EXEC_OUT_ATTEMPT_CAUSE	The cause of the procedure for HO Execution. – CS – CA – Service or Priority-triggered Inter-frequency Handover (SPIFHO) – ATO
INTERNAL_PROC_HO_EXEC_S1_OUT	EVENT_PARAM_PROC_HO_EXEC_OUT_ATTEMPT_CAUSE	The cause of the procedure for HO Execution. – CS

Event	Event Parameter	Description
		– CA – Service or Priority-triggered Inter-frequency Handover (SPIFHO) – ATO
INTERNAL_PROC_HO_EXEC_X2_IN	EVENT_ARRAY_HO_IN_PREP_ERAB_RESULT	Indicates handover reject due to offload admission.
INTERNAL_PROC_HO_EXEC_S1_IN	EVENT_ARRAY_HO_IN_PREP_ERAB_RESULT	Indicates handover reject due to offload admission.
INTERNAL_EVENT_UE_MOBILITY_EVAL	EVENT_PARAM_MOBILITY_TRIGGER	Indicates the trigger for performing a mobility evaluation. – CS – CA – Service or Priority-triggered Inter-frequency Handover (SPIFHO) – ATO
INTERNAL_EVENT_MEAS_CONFIG_A5	EVENT_PARAM_EUTRA_THRESHOLD1	The RSRP threshold value is 1. See 3GPP TS 36.331.
	EVENT_PARAM_EUTRA_THRESHOLD2	The RSRP threshold value is 2. See 3GPP TS 36.331.
	EVENT_PARAM_HYSTERESIS	The hysteresis value for the event A5 measurement. See 3GPP TS 36.331. **Note:** It is recommended to be set to ReportConfigEUtraInterFreqLb.hysteresisA5.
	EVENT_PARAM_MEAS_REQUESTER	Indicates the function that requested the UE measurement.

7 Activate Inter-Frequency Offload

Prerequisites

- The license key is installed in the node.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.
- Either the feature Coverage-Triggered Inter-Frequency Handover or Intra-LTE Inter-Mode Handover is activated.
- The cell relations where offload is desired must be configured to allow offload actions, by setting the relevant `EUtranCellRelation.loadBalancing` attributes to `OFFLOAD` or `IFO_AND_IFLB`.

  Note: The `IFO_AND_IFLB` value can only be used if Preferential Traffic Management is enabled and configured.

Steps

1. Set the `FeatureState.featureState` attribute to `ACTIVATED` in the `FeatureState=CXC4011557` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

8 Deactivate Inter-Frequency Offload

Prerequisites

CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `FeatureState.featureState` attribute to `DEACTIVATED` in the `FeatureState=CXC4011557` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

9 Engineering Guidelines for Inter-Frequency Offload

9.1 Configuration of Load Balancing Relation

> Load balancing relations are configured in the source cell towards each target cell, by setting the `EUtranCellRelation.loadBalancing` attribute.

Table 7 Configuration of Load Balancing Relation

MO Attribute	Value	Description
`EUtranCellRelation.loadBalancing`	`NOT_ALLOWED`	Setting the `EUtranCellRelation.loadBalancing` attribute to `NOT_ALLOWED` deactivates Inter-Frequency Load Balancing and Inter-Frequency Offload to the related target cell, in one direction.
	`ALLOWED`	Setting the `EUtranCellRelation.loadBalancing` attribute to `ALLOWED` activates Inter-Frequency Load Balancing to the related target cell, in one direction.
	`OFFLOAD`	Setting the `EUtranCellRelation.loadBalancing` attribute to `OFFLOAD` activates Inter-Frequency Offload to the related target cell, in one direction.
	`IFO_AND_IFLB`	Setting the `EUtranCellRelation.loadBalancing` attribute to `IFO_AND_IFLB` activates both Inter-Frequency Load Balancing and Inter-Frequency Offload to the related target cell, in one direction. **Note:** The `IFO_AND_IFLB` value can only be used if Preferential Traffic Management is enabled and configured.

The `EUtranCellRelation.loadBalancing` attribute can be configured in one direction, from the source cell to the target cell.

To apply the same configuration in both directions between two cells, `EUtranCellRelation.loadBalancing` must be set to the same value in both directions. This allows both cells to act as source cells to each other.

The `EUtranCellRelation.loadBalancing` attribute also controls the exchange of load information. When set to `ALLOWED` or `IFO_AND_IFLB`, the source cell requests the target cell to provide traffic load reports.

Other mobility functionality is not affected by the `EUtranCellRelation.loadBalancing` attribute.

Note: Inter-Frequency Load Balancing does not consider and may override restriction of connected mode mobility at coverage-triggered session continuity. Coverage-Triggered Inter-Frequency Session Continuity describes the configuration of the connected mode mobility restrictions.

The `EUtranCellRelation.loadBalancing` attribute can be determined and configured automatically by the Best Neighbor Relations for Intra-LTE Load Management feature.

9.2 Configuration of Subscription Parameters

The operation of the Inter-Frequency Load Balancing feature requires that QCI subscription quanta and cell subscription capacity are configured. QCI subscription quanta can be the `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` attribute. The cell subscription capacity can be the `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` attribute.

Configuration of QCI Subscription Quanta

The `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` attributes are configured in proportion to the typical traffic load an E-RAB of the specific QoS class represents in LTE RAN at high load. The general recommendation is to choose values for these attributes in proportion to the expected downlink bit rates (in kbps) under those conditions.

Note: The `qciSubscriptionQuanta` attribute does not influence the scheduling of the E-RAB, nor is it used for admission control. It is used solely to express the typical traffic load an E-RAB of the specific QoS class imposes on LTE RAN.

In case of combination with the Automated Cell Capacity Estimation feature, the principle of configuring `qciSubscriptionQuanta` is changed. For more information, see the document Automated Cell Capacity Estimation.

For a GBR type of E-RAB, the `qciSubscriptionQuanta` parameter is chosen in proportion to the typical guaranteed downlink bit rate (in kbps) for the particular QCI.

Table 8 Recommendations to Configure qciSubscriptionQuanta for GBR E-RABs

QCI	Recommendation
1 (conversational voice)	The value depends on the typical voice codec in use.
Other (2..4, ...)	The value depends on the type of streaming media.

For a non-GBR type of E-RAB, the `qciSubscriptionQuanta` parameter is chosen in proportion to what is considered an acceptable downlink bit rate (in kbps) for a UE in connected mode in LTE RAN at high load.

Table 9 Recommendations to Configure qciSubscriptionQuanta for Non-GBR E-RABs

Type of E-RAB	Recommendation
Default non-GBR types	The default non-GBR types of E-RAB are activated whenever the UE enters connected mode, irrespective if they are going to be used. Conservative values are chosen, reflecting an expected low or moderate duty rate on these E-RABs. In particular, the inactivity timer configured in the cell must be considered. A reduction of the inactivity timer may reduce the UE time spent in connected mode, whilst the UE data consumption is unaffected. The value of the `qciSubscriptionQuanta` attribute is increased to reflect the typical increase in average bit rate. If the inactivity timer is increased, the value of `qciSubscriptionQuanta` should be decreased. Note: The document Radio Bearer Service describes the configuration of the inactivity timer.
IMS signaling (default QCI = 5)	The expected bit rate is typically low (not exceeding 3 kbps).
Default ISP E-RAB	A suitable value can be based on the typical UE data consumption. This consumption is over a longer period of time in relation to the time spent in connected mode over the same period of time. The value has to focus on the conditions in LTE RAN at high load.
On-demand non-GBR type	Values have to be chosen based on the typical minimum QoS requirements of the intended applications.

The default E-RAB can be distinguished on different QCIs for users of separate categories, for example, for users with different types of UE devices. It offers a way to differentiate the `qciSubscriptionQuanta` attribute values among users of different categories. This can be useful if there are user categories with widely different QoS demand.

Note: A separation of user categories on separate QCIs requires support from the core network. This is not a feature of LTE RAN.

Configuration of Cell Subscription Capacity

At high load in LTE RAN, the `cellSubscriptionCapacity` (in MOs `EUtranCellFDD` or `EUtranCellTDD`) is configured in proportion to the achievable traffic capacity in the cell.

The achievable traffic capacity is typically proportional to the RF bandwidth of the cell, but can also depend on other factors, including the following:

- Inter-site distances and the presence of radio interference from neighbor cells.
- How the cell antenna is configured. For example, whether MIMO is applied.
- How the UE group in the cell is distributed between the cell center and the cell edge.

Note: If Inter-Frequency Load Balancing is used in combination with the features Automated Cell Capacity Estimation and Limited-Uplink-Aware IFLB, the function of the `cellSubscriptionCapacity` attribute can be replaced by these features. This determines and configures the cell capacity automatically. In this case, the principle of configuring `qciSubscriptionQuanta` is changed. For more information about the configuration, see the documents Automated Cell Capacity Estimation and Limited-Uplink-Aware IFLB.

It is recommended to choose a value of the attribute `cellSubscriptionCapacity` in proportion to the achievable downlink bit rate (in kbps) in the cell, considering the factors affecting the capacity. In a typical LTE RAN deployment, without advanced antenna configurations, the achievable spectrum efficiency at high load is usually in the proximity of 2 Mbit/MHz (bit rate for each RF bandwidth).

The amount of E-RABs the cell can accommodate is ultimately a trade-off with respect to the achievable QoS level. Increasing the amount of E-RABs has the following effects:

- Reduced throughput and increased packet delay for individual (non-GBR) E-RABs
- Increased subscription ratio, with a given configuration of the `qciSubscriptionQuanta` and `cellSubscriptionCapacity` attributes
- Declined user experience, yet a still operable cell

Therefore, the subscription ratio can be used as an indicator of the user experience in the cell.

The Inter-Frequency Load Balancing feature can handle load balancing also at subscription ratio levels greater than 1.0 (normally conceived as a mark of high load). The upper limit of the subscription ratio is 16. If the upper limit is reached, the load balancing can stall, because traffic load differences between cells above the limit are not perceptible. The value of the `cellSubscriptionCapacity` attribute should be chosen large enough to keep a margin with respect to this limitation.

9.3 Tuning of Subscription Parameters

In addition to the basic recommendations to configure load balancing, further tuning of the QCI subscription quanta and cell subscription capacity is possible to improve the load balancing behavior.

The QCI subscription quanta can be the `QciProfilePredefined.qciSubscriptionQuanta` or `QciProfileOperatorDefined.qciSubscriptionQuanta` MO attribute. The cell subscription capacity can be the `EUtranCellFDD.cellSubscriptionCapacity` or `EUtranCellTDD.cellSubscriptionCapacity` MO attribute.

For example, if a high E-RAB density of certain QoS class tends to cause the QoS level to decline. Then an increase of the corresponding `qciSubscriptionQuanta` attribute can help to reduce the number of UEs in the cells where that type of E-RAB is frequent.

The load balancing process does not consider the absolute level of `qciSubscriptionQuanta` and `cellSubscriptionCapacity`. The relations are important between these parameters.

Relation Between qciSubscriptionQuanta Values for Different QCIs

The relations between `qciSubscriptionQuanta` values for different QCIs determine how UEs with different E-RAB configuration can be exchanged in the load balancing process.

Increasing the `qciSubscriptionQuanta` attribute value for one QoS class tends to relocate UEs from cells with high density of this type of E-RAB to cells with lower density. The opposite is also true.

Relation Between cellSubscriptionCapacity Values of Different Cells

The relations between the `cellSubscriptionCapacity` values of different cells determine how the UE group is divided between cells where load balancing is allowed.

Increasing the value of the `cellSubscriptionCapacity` attribute for one cell attracts UEs to that cell. The opposite is also true.

Relation Between cellSubscriptionCapacity and qciSubscriptionQuanta Values

The overall relation between the values of `cellSubscriptionCapacity` and `qciSubscriptionQuanta` attributes determines the absolute level of the subscription ratio in the cell.

Increasing the overall ratio between the values of attributes `cellSubscriptionCapacity` and `qciSubscriptionQuanta` requires a higher number of UEs in the cell to reach the same subscription ratio.

9.4 Uplink-Heavy UE Detection Configuration

> Uplink-heavy UE detection can be enabled independently for the feature to improve the UE selection process for load-balancing mobility actions.

The following attributes must be set to `true` to enable uplink-heavy UE detection for load-balancing mobility actions:

- `EUtranCellFDD.lbUlHeavyUeDetectEnabled`
- `EUtranCellTDD.lbUlHeavyUeDetectEnabled`

The following can be used to configure uplink-heavy UE detection:

- `ENodeBFunction.ulHeavyUeDetectTime`
- `ENodeBFunction.ulHeavyUeBsrThresh`
- `ENodeBFunction.ulHeavyUeBsrHyst`

The attributes that configure uplink-heavy UE detection are the same attributes used by the Uplink-Traffic-Triggered Mobility and the Data-Aware Carrier Aggregation features. Therefore, changing these attributes also affects their uplink-heavy detection function.

Appendix A: Feature Change History

> This section lists changes that affected this feature and the impact it had on the network.

Appendix A.a: 19.Q1: Preferential Traffic Management

> The Preferential Traffic Management solution enables the operator to control user access to priority and non-priority bands. With Preferential Traffic Management, it is possible to define up to 12 subscriber groups denoted by a list of SPIDs and PLMN with different selection priority for Inter-Frequency Offload and Admission-Triggered Offload.

Feature Name:	System Improvements in LTE RAN L19.Q1
Feature Identity:	20/FAJ 121 1447
Value Package Name:	LTE Base Package
Value Package Identity:	FAJ 801 0400
Node Type:	Baseband Radio Node, DU Radio Node
Access Type:	LTE

The Preferential Traffic Management solution offers the operator the possibility to efficiently use an LTE network, deployed on several frequency bands, with one frequency band used primarily for priority users. The solution enables the following:

- To use the priority band for both priority and non-priority users as long as co-existence rules allow it.
- To use the priority band exclusively for priority users in the intended traffic situations.
- To use the non-priority bands for both non-priority and priority users, both as coverage bands and bands to offload priority traffic when the priority band is overloaded.
- To define how Preferential Traffic Management functions treat specific UEs by using subscriber groups.

The load management policies used depend on the current traffic load level. Traffic load states in the solution are defined by configurable load threshold parameters.

The solution is developed for existing commercial RAN infrastructure with a designated band used as priority band, using an MOCN architecture. The solution is achieved by a collection of extensions in existing licensed features, configured to work together.

For more detailed information on Preferential Traffic Management, see Preferential Traffic Management Guidelines.

Capacity and Performance

The capacity on priority bands is limited for non-priority users and changes with network congestion level. This is because of the need to provide uninterrupted connectivity for priority users. During network congestion, an increased drop in mobility success rate towards priority cells is expected.

Operation

The Preferential Traffic Management solution is activated on node level using the `PtmFunction.ptmEnabled` attribute.

On the eNodeB, a maximum of four cell profiles can be configured, but a cell refers to only one predefined profile. This controls which Preferential Traffic Management actions are triggered on the cell. Each cell profile can be configured using the `PtmCellProfile.cellType` attribute with the following settings:

- `PRIORITY`
- `NON_PRIORITY`

UE selection for Admission-Triggered Offload and Inter-Frequency Offload works only if the `PtmCellProfile.cellType` attribute is set to `PRIORITY`.

The function introduces the following new MO classes:

- PtmAtoConfig
- PtmCellProfile
- PtmFunction
- PtmIflbConfig
- PtmIfoConfig
- PtmResOpUseConfig
- PtmStmConfig
- PtmSubscriberGroup

This function is controlled by the following new parameters:

- PtmAtoConfig.minSuccRateThreshold
- PtmAtoConfig.ptmSubscriberGroupRef
- PtmCellProfile.cellType
- PtmFunction.ptmEnabled
- PtmIflbConfig.stopIncomingIflbThreshold
- PtmIflbConfig.stopOutgoingIflbEnabled
- PtmIfoConfig.ptmSubscriberGroupRef
- PtmResOpUseConfig.ptmSubscriberGroupRef
- PtmResOpUseConfig.resMsrUsageThreshold
- PtmResOpUseConfig.resSRatioThreshold
- PtmResOpUseConfig.unresMsrUsageThreshold
- PtmResOpUseConfig.unresSRatioThreshold
- PtmStmConfig.inhibitImpInterval
- PtmStmConfig.ptmSubscriberGroupRef
- PtmSubscriberGroup.plmn
- PtmSubscriberGroup.spidList

Handover of UEs with certain PLMN IDs and SPID to a target cell can be blocked. This occurs when the PLMN ID of the UE is "reserved for operator use" in the target cell. It can be reserved, because the priority cell has a very high traffic load—subscription ratio or MSR usage. It is also required that the SPID of the UE is configured in the PtmSubscriberGroup MO as the PtmStmConfig MO refers to that group.

The following features are affected by the solution:

Table 10 Affected Features

Feature	Description
Admission-Triggered Offload, FAJ 121 3100	Admission-Triggered Offload can be enabled per cell in a node to given target frequencies for UEs configured with a specific SPID.
Best Neighbor Relations for Intra-LTE Load Management, FAJ 121 3028	New value IFO_AND_IFLBEUtranFreqRelation.lbBnrPolicy to allow automatic configuration for offload and load balancing actions.
Inter-Frequency Load Balancing, FAJ 121 3009	Inter-frequency load balancing can be stopped for some cells—to priority cells with high traffic load from non-priority cells. This allows the priority band to be used primarily to serve priority users, and ensures access for priority UEs in high traffic load situations. is introduced for attribute
Inter-Frequency Offload, FAJ 121 3061	Inter-Frequency Offload can be enabled per cell in a node to given target frequencies for UEs configured with a specific SPID.

Feature	Description
Subscriber Triggered Mobility, FAJ 121 1788, RFPM function	Idle mode prioritization can be inhibited between some cells—to priority cells with very high traffic load from non-priority cells. This ensures that priority cells are not overloaded, so priority UEs have access in very high load traffic situations. The `offloadAllowed` struct member in FreqPrioEUTRA struct is introduced to allow or disallow inter-frequency offload for UEs with a specific SPID to a target frequency.

The following attributes are affected by the solution:

Table 11 Affected Parameters

Parameter	Description
`EUtranFreqRelation.lbBnrPolicy`	New value `IFO_AND_IFLB` is is introduced for attribute `EUtranFreqRelation.lbBnrPolicy` to allow automatic configuration for offload and load balancing actions. **Note:** The `IFO_AND_IFLB` value can only be used if Preferential Traffic Management is enabled and configured.
`EUtranCellRelation.loadBalancing`	New value `IFO_AND_IFLB` is introduced for the attribute to allow Inter-Frequency Offload and Inter-Frequency Load Balancing for the cell. **Note:** The `IFO_AND_IFLB` value can only be used if Preferential Traffic Management is enabled and configured.
`loadBalancingAllowed`	A struct member in FreqPrioEUTRA struct of attribute `RATFreqPrio.freqPrioListEUTRA`. Offloading behavior can be controlled by `offloadAllowed` struct member in FreqPrioEUTRA struct, if `loadBalancingAllowed` is set to FALSE.

Interfaces

This change affects the following interfaces:

- X2

  New private information element for traffic load state of a cell is added to cell resource status report messages.

Hardware

No special hardware requirements.

Other Network Elements

To fully benefit from the change, the node requires support from the following network elements:

- ENM

  The enhancement requires ENM version 18.16 or later to be used.

Appendix A.b: 22.Q4: Reserved Cell for Operator Use Improvement

> This system improvement enables priority UEs to select and access priority cells in high traffic load when the priority core network is not available.

Feature Name	System Improvements in L22.Q4
Feature Identity	35/FAJ 121 1447
Value Package Name	LTE Base Package

Value Package Identity	FAJ 801 0400
Node Type	Baseband Radio Node
Access Type	LTE
Impact at Upgrade	Optional

Summary and Benefits

This system improvement enhances accessibility for priority UEs as part of the Preferential Traffic Management solution. It enables priority UEs that utilize non-priority equivalent home PLMN to connect to priority bands and to fully utilize priority cell resources under the following circumstances:

- The priority cells are in a high traffic load.
- The priority PLMN core network is not available due to S1-C connection failure.

Capacity and Performance

This system improvement can be observed with the following PM counters:

- `EUtranCellFDD.pmPtmCellReservedMsr`
- `EUtranCellTDD.pmPtmCellReservedMsr`
- `EUtranCellFDD.pmPtmCellReservedSubRatio`
- `EUtranCellTDD.pmPtmCellReservedSubRatio`

Operation

This system improvement introduces the `PtmResOpUseConfig.suspResOpUsePlmnList` attribute. This attribute can suspend dynamic PLMN reservation during S1-C connection failure. The operator can select priority PLMNs for which the cell reservation is suspended.

If the `PtmResOpUseConfig.suspResOpUsePlmnList` attribute is empty, which is the default value, any of the priority PLMNs can suspend dynamic PLMN reservation for cells during S1-C connection failure.

This system improvement introduces the following PM events:

Table 12 Introduced PM Events

PM Event	Description
`INTERNAL_EVENT_PTM_CELL_RESERVED_START`	Triggered when the load is above threshold or the S1-C connection to the priority PLMN is recovered. It indicates that the cell is reserved for operator use.
`INTERNAL_EVENT_PTM_CELL_RESERVED_STOP`	Triggered when the load is below threshold or the S1-C connection to the priority PLMN failed. It indicates that the cell is no longer reserved for operator use.
`INTERNAL_EVENT_PTM_CELL_RESERVED_SUSP_START`	Triggered when the load is above threshold or the S1-C connection to the priority PLMN failed. It indicates that the cell reservation for operator use is suspended.
`INTERNAL_EVENT_PTM_CELL_RESERVED_SUSP_STOP`	Triggered when the load is below threshold or the S1-C connection to the priority PLMN is recovered. It indicates that the suspension of cell reservation for operator use is over.

Interfaces

No impact.

Hardware

No specific hardware requirements.

Other Network Elements

It is recommended to use ENM version 22.4 or later to support the MOM changes introduced by this system improvement. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

Related Information
Preferential Traffic Management Guidelines
Admission-Triggered Offload
Best Neighbor Relations for Intra-LTE Load Management
Inter-Frequency Load Balancing
Inter-Frequency Offload
Subscriber Triggered Mobility
Shared LTE RAN

Appendix A.c: 24.Q2.0: Uplink-Heavy UE Exclusion for Inter-Frequency Offload

> This feature enhancement enables the exclusion of UEs marked as uplink-heavy from load-balancing actions.

Feature Name	Inter-Frequency Offload
Feature Identity	FAJ 121 3061
Value Package Name	Inter-Vendor Load Management
Value Package Identity	FAJ 801 0416
Node Type	Baseband Radio Node
Access Type	LTE
Impact at Upgrade	Mandatory

Summary and Benefits

This feature enhancement introduces a mechanism to detect uplink-heavy UEs. If a UE is identified as uplink-heavy, it can be excluded from load-balancing mobility actions in FDD and TDD cells.

This feature enhancement has the following benefits:

- It is possible to detect uplink-heavy UEs for the Inter-Frequency Offload feature.
- The UE selection process is improved for load-balancing actions.
- Ping-pong effect between cells for uplink-heavy UEs is reduced when the Inter-Frequency Offload and the Uplink-Traffic-Triggered Mobility features are active at the same time in the network.

Capacity and Performance

The feature enhancement decreases the number of uplink-heavy UE handovers triggered by the Uplink-Traffic-Triggered Mobility feature. This can be observed with the following PM counters in the source cells where the Uplink-Traffic-Triggered Mobility feature is active:

- `EUtranCellFDD.pmCellHoPrepAttLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoPrepAttLteInterFUlTraffic`
- `EUtranCellFDD.pmCellHoPrepSuccLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoPrepSuccLteInterFUlTraffic`
- `EUtranCellFDD.pmCellHoExeAttLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoExeAttLteInterFUlTraffic`
- `EUtranCellFDD.pmCellHoExeSuccLteInterFUlTraffic`
- `EUtranCellTDD.pmCellHoExeSuccLteInterFUlTraffic`
- `EUtranCellRelation.pmHoPrepAttLteInterFUlTraffic`
- `EUtranCellRelation.pmHoPrepSuccLteInterFUlTraffic`

- `EUtranCellRelation.pmHoExeAttLteInterFUlTraffic`
- `EUtranCellRelation.pmHoExeSuccLteInterFUlTraffic`

Operation

The feature enhancement introduces attributes to control the detection of uplink-heavy UEs. When the following attributes are set to `true`, UEs marked as uplink-heavy are excluded from load-balancing mobility actions:

- `EUtranCellFDD.lbUlHeavyUeDetectEnabled`
- `EUtranCellTDD.lbUlHeavyUeDetectEnabled`

The following existing attributes are updated to control the detection of uplink-heavy UEs when the Inter-Frequency Offload feature is not active:

- `EUtranCellFDD.ulHeavyUeDetectEnabled`
- `EUtranCellTDD.ulHeavyUeDetectEnabled`

Note: These attributes are only applicable when at least one of the following features is active:

- Uplink-Traffic-Triggered Mobility
- Data-Aware Carrier Aggregation Configuration

The following legacy attributes are used for specifying the configurations of uplink-heavy UE detection for load balancing:

- `ENodeBFunction.ulHeavyUeDetectTime`
- `ENodeBFunction.ulHeavyUeBsrThresh`
- `ENodeBFunction.ulHeavyUeBsrHyst`

When uplink-heavy UE detection for load balancing is enabled in the cell, the mechanism is applied to UEs that connect to the LTE cell through the following processes:

- Incoming handover
- RRC initial setup
- RRC connection re-establishment

Interfaces

No impact.

Hardware

No specific hardware requirements.

Other Network Elements

It is recommended to use ENM version 24.2 or later to support the MOM changes introduced by this feature enhancement. If the Release Independence Manager application is used, the MOM changes are supported on earlier ENM versions also.

Related Information
Data-Aware Carrier Aggregation Configuration
Inter-Frequency Offload
Uplink-Traffic-Triggered Mobility

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